

AMRITA KAUR BEDI

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EDUCATION

University of California, Irvine

June 2028

Bachelor of Science in Biomedical Engineering | Minor in Bioinformatics | Minor in Innovation & Entrepreneurship

Relevant Courses – BME 60A (Arduino), BME 60B (MATLAB), ICS 33 (Python), BME 60C (SolidWorks - upcoming)

Cumulative GPA: 3.948 | Dean's Honor List – 4 Quarters

EXPERIENCE

Applied Medical Laparoscopic Trainer Lab Module Development

January 2026 - Present

Team Leader

- Design and execute bench-top verification testing protocols and custom test fixtures to quantify laparoscopic tool-tissue interaction forces using Arduino-based data acquisition systems
- Apply ISO surgical instrument standards and FDA 510(k) design control principles to evaluate medical device performance and safety margins
- Author and refine standard operating procedures (SOPs) and support quantitative data analysis for an ABET-accredited biomedical engineering laboratory course, with materials structured for future dissemination

Novel Suture Project

May 2025 – Present

Fabrication Lead

- Designed, fabricated, and calibrated a custom PCB-based pressure sensor (**range 0–183 kPa, $R^2 = 0.997$**) for biomedical force monitoring
- Conducted prototype hardware validation using MATLAB, achieving **0.20 kPa** resolution and **26.5 dB** SNR, with focus on repeatability, accuracy, and process optimization
- Authored and implemented SOPs for fabrication, testing, and design verification under quality control standards
- Presented at 2025 National BMES Conference and UCI Undergraduate Research Opportunities Program Symposium

Downing Lab at UCI

April 2025 – Present

Data Analytics Lead

- Investigated cytoskeletal changes in lung cancer cell lines following CSRP1 knockdown using immunofluorescence staining and quantitative data analysis (**$p < 0.0001$**)
- Measured **20+** cell morphology parameters using ImageJ, Python, and R to assess structure-function relationships
- Presented [findings](#) at the 2025 National BMES Conference

UltraScan Pro: Arduino-Based Ultrasonic Imaging System

September 2025 – December 2025

Team Leader

- Directed a 4-member engineering team to design and prototype a simplified [ultrasound-style ranging system](#)
- Integrated ultrasonic sensors, servo motor control, LCD interface, and real-time MATLAB visualization to generate a 180° spatial map of surrounding objects
- Developed interactive feedback features (LED indicators, motion alerts) to improve system usability and interpretability
- Delivered a final solution that placed **2nd out of 19 teams**

Remote Controlled Rover

September 2024 – December 2024

Team Leader

- Directed a 7-member team through full mechanical design and testing cycle of a remote-controlled rover
- Modelled assemblies in SolidWorks and enhanced steering performance by **20%** through Ackermann geometry optimization and iterative testing

EXTRACURRICULARS

Biomedical Engineering Society at UCI

January 2025 – Present

Project Coordinator

- Founded and currently manage the MedTech Founders Program, mentoring **20+** students through the full product development lifecycle from concept to prototype
- Organized **3+** technical workshops on SolidWorks, Arduino, and programming, improving members' design for manufacturability and rapid prototyping skills, impacting **50+** undergraduates
- Coordinated cross-functional events engaging **100+** students in hands-on engineering projects

SKILLS

Prototyping: SolidWorks (CAD), 3D Printing, PCB Design, Soldering, Laser Cutting, PDMS Fabrication, Rapid Prototyping

Testing & Validation: Bench Testing, Mechanical Testing, Design Verification, Experimental Design, Fixture Development, Root Cause Analysis

Programming & Data Analysis: Python, MATLAB, R, Arduino, Data Visualization, ImageJ, SQL

Product Development & Documentation: Technical Documentation, SOP Development, Design Validation, Cross-Functional Collaboration